RESEARCH PROJECT

Multi-Document RAG Evaluation Framework

RAG RankRAG Self-RAG MADAM-RAG RAGChecker

End-to-end benchmark pipeline grounded in 5 research papers

HotpotQA · RAMDocs · 3 LLM Backends · Faithfulness Analysis

The Problem

Multi-document question answering is fundamentally harder than single-passage retrieval

EXAMPLE MULTI-HOP QUESTION

"What is the nationality of the director of Spectre?"

📄 Doc 1: Spectre (film) → director = Sam Mendes +

📄 Doc 2: Sam Mendes → nationality = British → ✅ British

🔗

Multi-hop Reasoning

Answer requires chaining evidence across 2+ documents — no single doc contains the full answer

🎭

Misinformation in Corpus

Real corpora contain adversarial documents designed to mislead — model must suppress them

🌀

Hallucination Risk

LLMs generate plausible but unsupported text — hard to distinguish from grounded answers

Existing baselines:

What We Built

A unified, end-to-end benchmark — retrieval through faithfulness, all in one pipeline

📊 Benchmark Suite

- 2 datasets (HotpotQA, RAMDocs)
- 5 retrieval methods
- 3 LLM backends
- 400+ test queries

⚙️ Production Pipeline

- Retrieval → Reranking
- Multi-agent debate
- Rule-based + LLM generation
- Faithfulness checking

🔍 Failure Analysis

- Retrieval vs generation failures
- Hallucination detection
- Misinformation suppression

- Joint F1 tracking

Datasets

multi-hop reasoning

10 candidate docs/query · 200 test examples · bridge-entity questions

adversarial misinformation

Gold / Misinfo / Noise docs · 200 test examples · ambiguous entities

Metrics Tracked

Research Foundation

Each pipeline component is directly grounded in a peer-reviewed paper

Lewis et al., 2020

Retrieval-Augmented Generation — the foundational pattern

Architecture

Yu et al., 2024

LLM as pointwise reranker — score each doc independently

Reranking

Asai et al., 2023

Iterative retrieval with reflection tokens for adaptive queries

Retrieval

Yoran et al., 2024

Multi-agent debate — one agent per doc, arbitrator decides

Debate

Ru et al., 2024

Claim-level faithfulness checking + failure mode taxonomy

Eval

Paper → Component mapping: RAG

Self-RAG

RankRAG

MADAM-RAG

RAGChecker

Novel contribution: **single runnable pipeline**

Circuit breaker:

System Architecture

9-step pipeline from raw question to verified answer

Three Retrieval Strategies

Progressive complexity — each addresses a limitation of the previous

Baseline

BM25

Sparse keyword matching using TF-IDF weights. Fast, interpretable, strong baseline.

① Tokenize query
② Score docs by term overlap
③ Return Top-K by BM25 score

Limitation: misses semantic synonyms

Improved

Hybrid (BM25 + Dense)

Fuses sparse BM25 with dense semantic embeddings via Reciprocal Rank Fusion.

① BM25 score (sparse)
② Sentence-transformer embeddings
③ RRF fusion → re-ranked list

Gain: captures paraphrases + semantic similarity

Best

Iterative Hybrid

Two-pass retrieval inspired by Self-RAG: first pass extracts bridge entities for second query.

① Pass 1: Hybrid on original question
② Extract bridge entities from results
③ Pass 2: Hybrid on augmented query
④ Merge both result sets

Gain: finds both docs in multi-hop chains

ITERATIVE HYBRID — MULTI-HOP EXAMPLE

"What nationality is the Spectre director?" → Pass 1 retrieves Spectre article →

Extracts bridge entity "Sam Mendes" → Pass 2 retrieves Sam Mendes bio →

Both gold docs found ✓

Reranking & Multi-Agent Debate

RankRAG-style LLM scoring feeds directly into MADAM-RAG debate

① LLM Reranker (RankRAG)

For each retrieved document, the LLM is asked:

```
"On a scale 1–10, how relevant is
this document to the question?"
```

- Assigns relevance score per doc
- Re-orders before debate
- Deterministic fallback: original order

Impact: EM jumps from 0.595 → 0.74 with BM25 + LLM reranker

② Multi-Agent Debate (MADAM-RAG)

$Agent_1$
gold doc

$Agent_2$
misinfo doc

$Agent_3$
noise doc

$Agent_4$
gold doc

↓ weighted votes

⚖️ Arbitrator — final answer

Label weights: gold → ×2.0 misinfo → ×−1.5 noise → ×−0.5

Misinfo suppressed: **96.5%** on RAMDocs

Faithfulness & Failure Analysis

RAGChecker-style evaluation — two paths, always produces a classification

Path A — LLM Judge (when available)

LLM receives: question · predicted answer · gold answer · retrieved docs

Returns JSON:

```
{
  "faithfulness_score": 7,
  "hallucinated_claims": ["..."],
  "failure_mode": "generation_failure"
}
```

Score 0–10 → normalised to 0.0–1.0

Path B — Deterministic (always runs)

EM = 1 → correct

EM=0, gold docs retrieved → generation_failure

EM=0, gold docs missing → retrieval_failure

Circuit breaker: if LLM returns "unknown" → deterministic result used

~90%

of all failures are
retrieval failures

~10%

of all failures are
generation failures

9:1

retrieval : generation
failure ratio

Experimental Setup

Four experimental runs — each isolating a different pipeline stage

Run	Retrieval	Reranker	Generation	Faithfulness	Pur
Baseline	BM25 / Hybrid	None	Rule-based	Deterministic	No ren flo
LLM Rerank	BM25 / Hybrid / Iterative	Groq 8B	Rule-based	Deterministic	Re im
Haiku Best ★	BM25	Haiku	Rule-based	LLM judge	Be qu LL
LLM Generation	Hybrid / Iterative	Groq 8B	LLM debate	LLM judge	Fu pip

LLM Backends Compared

Local:

Free cloud:

Paid:

Scale

K values

Results — Retrieval Quality

HotpotQA · LLM Rerank run · higher = better

Method	Recall@5	MRR	Multi-doc HR
BM25 (baseline)	0.650	0.850	0.245
BM25 + Rerank	**0.750**	**0.875**	**0.385**
Hybrid + Rerank	0.768	0.926	0.425
Iterative Hybrid ★	**0.770**	**0.926**	**0.435**

LLM reranker alone: +10% Recall@5 over baseline BM25

Recall@5 comparison

MRR improvement

Results — Answer Quality

HotpotQA Exact Match across pipeline configurations

Configuration	EM (HotpotQA)	Answer F1	Halluc. Rate
Baseline (BM25)	0.595	0.602	—
BM25 + LLM Rerank	**0.740**	**0.745**	—
Hybrid + LLM Rerank	0.710	0.716	—
Haiku + BM25 (best) ★	**0.800**	**0.800**	**0%**
LLM Gen · BM25	0.220	0.359	36%
LLM Gen · Hybrid	0.180	0.347	28%
LLM Gen · Iterative	0.260	0.422	18%

RAMDocs: all methods ≈ 0.96 EM Misinfo suppressed 96.5%

Answer EM — HotpotQA

Key Surprise

LLM generation EM dropped 0.80 → 0.22 — LLMs generate *correct but unmatched* text. Rule-based extraction wins on short factoid answers.

LLM Backend Comparison

Same pipeline, three backends — Ollama 3B · Groq 8B · Claude Haiku

Configuration	Backend	EM (HotpotQA)	Faithfulness	Halluc. Rate	Cost
BM25 + Ollama rerank	Llama 3.2 3B · local	0.650	—	—	Free
BM25 + Groq rerank	Llama 3.1 8B · cloud	0.740	0.482	36%	Free tier
Iterative + Groq	Llama 3.1 8B · cloud	0.705	0.574	18%	Free tier
BM25 + Haiku rerank ★	Claude Haiku · API	**0.800**	**—**	**0%**	Paid

🖥️

- Runs fully local, no API
- Weakest reranking quality
- Good for offline/private use
- EM: 0.650 (near baseline)

☁️

- Free cloud tier (30 RPM)
- Strong reranker, good F1
- Rate limit → 3.5 hr run
- Best faithfulness: 0.574

✨

- Highest answer quality
- **0.800 EM** — best overall
- 0% hallucination (rule-based)
- Paid but fast

Key Insights

What the numbers tell us

9:1

90% of wrong answers are due to the right documents *not being retrieved* — not hallucination. Fix retrieval first.

+20%

Adding a LLM reranker (Haiku) jumped EM from 0.595 → 0.800 — the single highest-impact component in the pipeline.

↓64%

Full LLM debate generation dropped EM from 0.80 → 0.22. LLMs paraphrase answers — rule-based extraction is better for factoid QA.

96.5%

MADAM-RAG debate with label-weighted voting actively suppresses adversarial documents — not just observed, actively blocked.

Practical takeaway:

For factoid QA → best config is Iterative Hybrid + Haiku reranker + rule-based extraction

For free-form QA → LLM generation with Iterative Hybrid gives best faithfulness (0.574) and lowest hallucination (18%)

Conclusion

What We Built

End-to-end RAG pipeline combining 5 research papers into one unified, runnable system

Three retrieval methods, LLM reranker, multi-agent debate with misinformation suppression

Faithfulness checking with deterministic fallback — graceful degradation at every step

Benchmarked across 400 queries, 3 LLM backends, full failure-mode taxonomy

What We Proved

EM 0.800 on HotpotQA — competitive with published baselines

96.5% misinformation suppression on adversarial RAMDocs

Better retrieval beats better generation — **9:1 failure ratio**

Iterative Hybrid uniquely solves multi-hop retrieval via bridge-entity chaining

5 papers	9 pipeline steps	400 queries	3 LLM backends	EM 0.800 ★

Speaker notes